STAFF REVIEW · H446 096 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Stores current/intermediate calculation.

2. Stores current ACC into mailbox X.

3. Reads input into ACC, replacing its former value.

4. Sum0 (XOR); Carry1 (AND).

I Do — source worked steps

1. Follow the reasoning — FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

We Do — guided CHECK

1. Follow the reasoning
After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.

You Do — source mark schemes

1. ACC4 after first INP [1], FIRST4 after STA [1], ACC7 after second INP [1], ACC11 after ADD [2], output11 [1].

2. INP [1], STA N [1], LDA N [1], OUT [1], HLT [1], N DAT 0 [1].

3. LDA copies mailbox to ACC [2], STA copies ACC to mailbox [2], ADD adds mailbox value into ACC [2].

Exit Ticket — source answer

1. False. LDA does that; STA writes to the mailbox.